Quaternion Approximate Networks for Enhanced Image Classification and Oriented Object Detection

Bryce Grant[1] and Peng Wang[2]

***Abstract*— This paper introduces Quaternion Approximate Networks (QUAN), a novel deep learning framework that leverages quaternion algebra for rotation equivariant image classification and object detection. Quaternion networks are studied for their inherent rotation equivariance, enabling them to capture richer spatial relationships and geometric structures compared to real-valued convolutions. Unlike conventional quaternion neural networks that operate entirely in the quaternion domain, QUAN utilizes the Hamilton product to approximate quaternion convolution while maintaining real-valued operations. This approach preserves both semantic and geometric information while reducing computational overhead, allowing for efficient implementation within standard deep learning frameworks. Additionally, quaternion operations are extended to spatial attention mechanisms, and Independent Quaternion Batch Normalization is introduced to enhance feature characterization stability.**

The effectiveness of QUAN is evaluated in two experimental settings: image classification and rotated object detection. In classification tasks on CIFAR-10 and CIFAR-100, QUAN achieves higher accuracy with fewer parameters and faster convergence compared to existing convolution and quaternion-based models. For rotated object detection, performance is assessed on a real-world robotic assembly dataset, where a UR5 robot executes manipulation tasks with oriented bounding box outputs. By integrating quaternion-based representations into both classification and detection tasks, QUAN demonstrates improved parameter efficiency and rotation handling over standard Convolution Neural Networks (CNNs) and improved accuracy over existing quaternion CNNs. These results highlight its potential for deployment in resource-constrained robotic systems and provide a foundation for further exploration on larger benchmark datasets.

***Keywords*—** Robotic Assembly, Quaternion Networks, Oriented Object Detection, Robotic Perception

I. INTRODUCTION

Vision-based robotic assembly and manipulation tasks represent a critical challenge in modern robotics, which requires precise object detection, pose estimation, and spatial reasoning capabilities. Conventional computer vision approaches typically struggle with object orientation, a fundamental requirement for robotic interaction with real-world objects. Although deep learning has revolutionized object detection through frameworks such as YOLO [1], Faster R-CNN [2], and DETR [3], most approaches rely on axis-aligned bounding boxes that discard crucial orientation information. This limitation significantly affects performance in robotic applications where grasp planning and assembly operations are highly dependent on accurate object orientation [4].

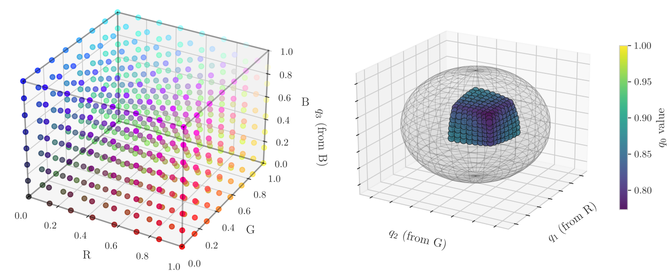

Fig. 1. Poincaré mapping from RGB color space to quaternion space

Recent advances in vision-language models and self-supervised learning have transformed the computer vision landscape. DINO [5] and other contrastive learning approaches [6] have demonstrated improved performance by learning from unlabeled data, while Mixture-of-Experts (MoE) architectures [7] achieve state-of-the-art results through adaptive computation pathways. However, these sophisticated models come with substantial computational requirements, often containing hundreds of millions of parameters. Despite their impressive general-purpose capabilities, they lack the inherent mathematical structures to efficiently represent and process rotational information, a critical requirement for robotic manipulation tasks.

Meanwhile, specialized approaches for oriented object detection [8], [9] have emerged to address rotation-aware perception needs. These methods typically use complex architectures and loss functions to estimate object orientation, but they often require substantial computational resources. This computational burden presents a significant barrier for deployment in resource-constrained robotic systems that require real-time performance.

Quaternion algebra offers a powerful mathematical framework for encoding and manipulating three-dimensional rotations [10], [11]. Originally developed by Hamilton in 1843, quaternions avoid the singularities associated with Euler angles (gimbal lock) and provide a more compact representation compared to rotation matrices. This mathematical elegance has inspired researchers to explore quaternion-based deep learning approaches. Gaudet and Maida [12] established foundational concepts in deep quaternion networks, introducing quaternion backpropagation and initialization

The author from [1] is with the Department of Electrical, Systems and Computer Engineering; Case Western Reserve University [2] is with Department of Mechanical and Aerospace Engineering, Case Western Reserve University, Cleveland, Ohio, USA

E-Mail: bag100@case.edu and pxw206@case.edu

Codes available at: https://github.com/bryceag11/QuatNet_OBB

This work is supported by the National Science Foundation under grant No. 2024614.

schemes for image classification, while Parcollet et al. [13] demonstrated the parameter efficiency of quaternion neural networks for audio processing tasks. Zhu et al. [14] further extended this to image denoising tasks, showing quaternion CNNs could achieve comparable performance to standard CNNs with significantly fewer parameters.

Despite these advances, existing quaternion neural networks face substantial implementation challenges. Pure quaternion networks require specialized mathematical operations that are poorly supported in mainstream deep learning frameworks, leading to significant computational overhead. Gomez and Gershenson [15] highlighted fundamental limitations such as the non-existence of bounded non-constant analytic quaternion functions, constraining the design of activation functions and complicating gradient computations. These practical limitations have prevented quaternion networks from being deployed in real-world robotic applications despite their theoretical advantages.

This paper introduces Quaternion Approximate Networks (QUAN), a novel approach that preserves the rotation equivariance benefits of quaternion representations while maintaining computational efficiency. Rather than performing operations in the quaternion domain directly, QUAN implements quaternion convolution through Hamilton product approximation using standard real-valued operations. This approach allows leveraging highly optimized tensor operations in existing deep learning frameworks while retaining the representational advantages of quaternion.

The key contributions are as follows:

- A quaternion approximation approach that implements Hamilton-based mixing using real-valued operations, maintaining rotation equivariance while significantly reducing computational complexity compared to previous quaternion neural networks.
- Novel quaternion-aware architectural components, including Independent Quaternion Batch Normalization (IQBN) and Quaternion Partial Spatial Attention (QC2PSA), that enhance model performance while preserving quaternion structure.
- State-of-the-art performance for quaternion networks on image classification tasks (CIFAR-10, CIFAR-100) with significantly reduced parameter counts compared to both standard CNNs and existing quaternion approaches.
- A systematic quaternion adaptation of YOLO architecture components (QC3k2, QSPPF, QC2PSA) for oriented bounding box detection, benchmarked on a custom robotic manipulation dataset.
- A novel orientation-aware loss function unifying quaternion angular metrics with traditional detection objectives, enabling end-to-end learning of both object location and orientation in quaternion space.

QUAN represents the first quaternion-inspired architecture that successfully bridges the gap between theoretical quaternion advantages and practical deployment needs for robotic perception systems. By combining parameter efficiency with rotation equivariance properties, this approach enables robust object detection and pose estimation for robotic manipulation tasks while maintaining suitable for real-time applications.

II. Related Works

A. General Object Detection and Pose Estimation

Object detection forms the foundation of many robotic vision systems. Landmark architectures like YOLO [1], Faster R-CNN [2], and DETR [3] have established new performance benchmarks but primarily focus on axis-aligned bounding boxes, which limit their utility for robotic manipulation tasks where orientation is crucial. For robotic applications, Wang et al. [4] developed DenseFusion, by fusing RGB and depth information, while Hodan et al. [16] introduced a benchmark for 6D object pose estimation in cluttered scenes while Kleeberger et al. [17] introduced a benchmark for industrial bin picking, highlighting the challenges of accurate orientation estimation in realistic settings. These approaches, while effective, often separate the detection and pose estimation steps rather than handling them in a unified framework.

Specialized methods for oriented object detection have emerged to address these limitations. Yang et al. [8] proposed R3Det, which refines feature maps for detecting rotated objects, while Xu et al. [9] introduced the gliding vertex mechanism to generate arbitrarily oriented bounding boxes. However, these approaches typically require complex architectural modifications and substantially increased computational resources compared to standard detectors.

B. Quaternion-Based Methods

Quaternions provide a compact and singularity-free representation for 3D rotations [10], [11]. Gaudet and Maida [12] pioneered quaternion neural networks by establishing fundamental operations, including weight initialization and backpropagation. Parcollet et al. [13] extended this work to recurrent networks and demonstrated parameter efficiency in speech processing tasks. Zhu et al. [14] applied quaternion convolutions to image processing, achieving comparable performance to standard CNNs with fewer parameters. Comminiello et al. [18] represented microphone in their spherical harmonics form, which enabled the use of quaternion networks for the detection of 3D sound events. Grassucci et al. [19] investigated quaternion generative adversarial networks, to obtain better FID scores than real-valued GANs. These works showed that quaternion CNNs could achieve comparable or even superior performance to standard CNNs with fewer parameters.

However, quaternion neural networks face significant implementation challenges. Gomez and Gershenson [15] identified limitations including higher computational complexity and the non-existence of bounded non-constant analytic quaternion functions, which constrains activation function design. These practical challenges have prevented widespread adoption despite theoretical advantages.

C. Quaternion Pose Estimation

Rotation equivariance is particularly valuable for robotic perception. Hsu et al. [20] introduced QuatNet, using quaternions for continuous head pose estimation with a multi-regression loss function. He et al. [21] introduced PVN3D, a deep point-wise 3D keypoints voting network for 6DoF pose estimation using quaternions for rotation representation. Despite these advances, no previous work has successfully applied quaternion-based approaches to oriented object detection in practical robotic applications, which is the focus of this paper.

III. Methodology

A. Quaternion Representation and Mapping

The quaternion number system extends complex numbers to four dimensions, represented as:

$$q = q_r + q_i\mathbf{i} + q_j\mathbf{j} + q_k\mathbf{k} \tag{1}$$

where $q_r, q_i, q_j, q_k \in \mathbb{R}$, and $\mathbf{i}, \mathbf{j}, \mathbf{k}$ are the fundamental quaternion units satisfying:

$$\mathbf{i}^2 = \mathbf{j}^2 = \mathbf{k}^2 = \mathbf{ijk} = -1 \tag{2}$$

For image processing applications, RGB data must be mapped into quaternion space. Several mapping strategies have been explored, as explained in the Appendix VI-A, with evaluation demonstrating that the Poincaré mapping [22], [23], which embeds the RGB cube within the unit quaternion sphere shown in Fig. 1, provided superior performance:

$$\begin{aligned}
\mathbf{norm} &= \sqrt{r^2 + g^2 + b^2} \\
r' &= \frac{r}{\mathbf{norm} + 1} \\
g' &= \frac{g}{\mathbf{norm} + 1} \\
b' &= \frac{b}{\mathbf{norm} + 1} \\
q_r &= \sqrt{1 - (r'^2 + g'^2 + b'^2)} \\
q &= [q_r, r', g', b']
\end{aligned} \tag{3}$$

The Poincaré mapping embeds RGB values into the into the Poincaré ball model of hyperbolic space, where the distance between points increases exponentially as they approach the boundary of the unit ball. This property allows the quaternion representation to better preserve hierarchical relationships in the color space compared to Euclidian embeddings.

B. Hamilton-Based Quaternion Approximation

The core innovation in QUAN is the approximation of quaternion convolution through real-valued operations. Traditional quaternion convolution applies the Hamilton product between quaternion-valued features and kernels:

$$\mathbf{W} \otimes \mathbf{X} = \begin{bmatrix} w_r & -w_i & -w_j & -w_k \\ w_i & w_r & -w_k & w_j \\ w_j & w_k & w_r & -w_i \\ w_k & -w_j & w_i & w_r \end{bmatrix} \begin{bmatrix} x_r \\ x_i \\ x_j \\ x_k \end{bmatrix} \tag{4}$$

Instead of implementing this as quaternion algebra directly, QUAN decomposes the operation into standard convolutions followed by a mixing step:1) four separate real-valued convolutions are applied to each quaternion component; and 2) the outputs are mixed according to the Hamilton product formula.

For an input tensor $X \in \mathbb{R}^{B\times C\times 4\times H\times W}$ and convolutional weights W, the separable QUAN convolution [24] is defined as:

$$\begin{aligned}
y_r &= W_r * x_r - W_i * x_i - W_j * x_j - W_k * x_k \\
y_i &= W_r * x_i + W_i * x_r + W_j * x_k - W_k * x_j \\
y_j &= W_r * x_j - W_i * x_k + W_j * x_r + W_k * x_i \\
y_k &= W_r * x_k + W_i * x_j - W_j * x_i + W_k * x_r
\end{aligned} \tag{5}$$

where $*$ denotes standard convolution. This approach maintains the mathematical properties of quaternion convolution while leveraging optimized tensor operations in existing deep learning frameworks. QUAN employs two split-type activation functions, Quaternion Sigmoid Linear Unit (QSiLU) and Quaternion Parametric Rectified Linear Unit (QPReLU), that operate separately on real and imaginary components [25]:

$$\text{QSiLU}(q) = (q_r, q_i \cdot \sigma(q_i), q_j \cdot \sigma(q_j), q_k \cdot \sigma(q_k)) \tag{6}$$

$$\text{QPReLU}(q) = (q_r, \text{PReLU}(q_i), \text{PReLU}(q_j), \text{PReLU}(q_k)) \tag{7}$$

C. Independent Quaternion Batch Normalization

To maintain quaternion structure while stabilizing training, QUAN introduces Independent Quaternion Batch Normalization (IQBN). Unlike previous quaternion normalization approaches that treat quaternions as unified entities, IQBN normalizes each component independently while preserving important cross-component relationships.

For quaternion feature $q = \{q_r, q_i, q_j, q_k\}$, with $B\times C\times Q\times H\times W$ shape, IQBN computes:

$$\begin{aligned}
\mu_c &= \mathbb{E}[X_c] \\
\sigma_c^2 &= \mathbb{E}[(X_c - \mu_c)^2] \\
\hat{x}_c &= \frac{x_c - \mu_c}{\sqrt{\sigma_c^2 + \epsilon}}
\end{aligned} \tag{8}$$

where $c \in \{r, i, j, k\}$ represents each quaternion component.

During training, running statistics are computed:

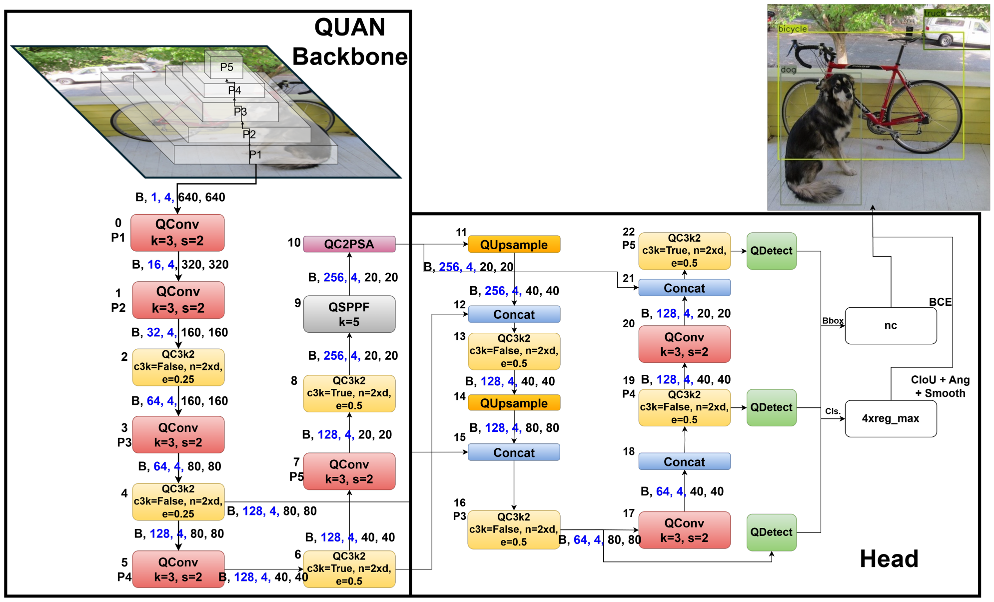

Fig. 2. QUAN architecture showing quaternion adaptations of key components: QConv, QC3k2, QSPPF, and QC2PSA

$$
\begin{aligned}
x_{\text{flat}} &= x.\text{reshape}(B, C, Q, -1) \\
\mu &= x_{\text{flat}}.\text{mean}([0, 3]) \\
\sigma^2 &= x_{\text{flat}}.\text{var}([0, 3]) \\
x_{\text{norm}} &= \frac{x - \mu.\text{view}(1, C, Q, 1, 1)}{\sqrt{\sigma^2.\text{view}(1, C, Q, 1, 1) + \epsilon}} \\
\text{Output} &= x_{\text{norm}} \cdot \gamma + \beta
\end{aligned} \tag{9}
$$

These are maintained for efficient inference:

$$
\begin{aligned}
x_{\text{norm}} &= \frac{x - \mu_{\text{running}}}{\sqrt{\sigma^2_{\text{running}} + \epsilon}} \\
\text{Output} &= x_{\text{norm}} \cdot \gamma + \beta
\end{aligned} \tag{10}
$$

D. Quaternion Architectural Components

As a novel framework that integrates quaternion approximation with IQBN, QUAN can be leveraged to renovate most CNNs and quaternion CNNs, with necessary modifications to maintain quaternion operations throughout the network structure. In this section, YOLOv11 [1] is renovated with QUAN, and the following modifications, illustrated in Fig. 2, were made besides replacing standard convolution with approximated quaternion convolution:

1) Quaternion Cross Stage Partial with kernel size 2 (QC3k2): The C3k2 module balances multi-scale features and gradient flow in YOLO architectures. The quaternion adaptation, QC3k2, maintains the partial cross-stage design while adapting convolutions to quaternion operations:

$$\text{QC3k2}(x) = \text{QConv}\left(\text{Concat}\left(\text{QBottleneck}(x_1), x_2\right)\right) \tag{11}$$

where x_1 and x_2 are the split components of the input feature map.

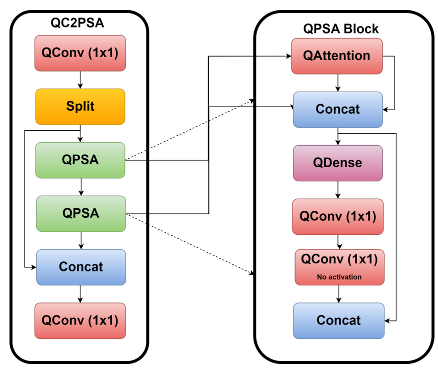

Fig. 3. Quaternion Partial Spatial Attention

2) Quaternion Spatial Pyramid Pooling (QSPPF): SPPF efficiently handles multi-scale features through pyramid pooling. In QSPPF, the quaternion structure is preserved during pooling operations by applying pooling operations independently to each component followed by Hamilton-based mixing. This approach maintains rotation equivariance while capturing multi-scale information.

3) Quaternion Partial Spatial Attention (QC2PSA): C2PSA enhances the network's ability to focus on relevant spatial regions while maintaining quaternion structure. The module splits input features, applies quaternion attention to one branch, and recombines them. The quaternion attention mechanism computes spatial attention weights while respecting quaternion algebra, ensuring that rotational information is preserved. The module implements the operation shown in Fig. 3. where QPSA is the quaternion adaptation of partial spatial attention, allowing the network to focus on spatially relevant features while preserving quaternion relationships.

E. Loss Functions for Object Detection and Orientation

For oriented object detection, QUAN employs specialized loss terms that account for both object localization, orientation, and classification:

$$\mathcal{L}_{\text{total}} = \mathcal{L}_{\text{cls}} + \mathcal{L}_{\text{CIoU}} + \lambda_1 \mathcal{L}_{\text{angular}} + \lambda_2 \mathcal{L}_{\text{reg}} + \lambda_3 \mathcal{L}_{\text{smooth}} \quad (12)$$

where:

- $\mathcal{L}_{\text{cls}} = -\sum_{c=1}^{C} y_c \log(\hat{y}_c) + (1-y_c)\log(1-\hat{y}_c)$ - the classification loss
- $\mathcal{L}_{\text{CIoU}} = 1 - \text{IoU} + \frac{\rho^2(\mathbf{b}, \mathbf{b}^2 gt)}{c^2} + \alpha v$ - the complete IoU loss for bounding box regression
- $\mathcal{L}_{\text{angular}} = 2\arccos(|\langle q_{\text{pred}}, q_{\text{target}} \rangle|)$ - the quaternion angular loss computes the geodesic angular distance between predicted and ground truth quaternion orientations, accounting for the double cover property
- $\mathcal{L}_{\text{reg}} = \|q\|_2 - 1^2$ - the quaternion regularization loss enforces unit quaternion constraints to maintain valid rotation representations
- $\mathcal{L}_{\text{smooth}} = \arccos(|\langle q_i, q_{i+1} \rangle|)$ - the orientation smoothness loss encourages consistency between sequential predictions to reduce jitter

This integrated loss formulation addresses the fundamental challenge of simultaneously optimizing detection accuracy and orientation precision, with the quaternion components specifically designed to preserve the geometric constraints essential for rotation representation in 3D space.

F. Implementation Details

QUAN is implemented in PyTorch, leveraging the framework's efficient tensor operations. To maximize computational efficiency, the quaternion operations are decomposed into standard tensor operations rather than implemented as custom quaternion algebra. This approach allows QUAN to benefit from hardware accelerations optimized for standard deep learning operations while maintaining the mathematical properties of quaternions.

The implementation defines custom modules for QConv2D, QDense, IQBN, and quaternion activation functions, all of which internally use standard PyTorch operations. This ensures compatibility with existing optimization techniques and deployment workflows.

IV. Experiments

This section presents a comprehensive evaluation of QUAN across image classification and oriented object detection tasks. All experiments were conducted on a single NVIDIA RTX 4090 GPU, ensuring a consistent hardware platform for fair comparison. The experiments are designed to evaluate three key aspects: (1) parameter efficiency compared to standard CNNs and existing quaternion networks, (2) rotation equivariance properties essential for robotic manipulation, and (3) real-world applicability in robotic assembly tasks.

A. Datasets

1) Image Classification Datasets: For image classification, CIFAR-10 and CIFAR-100 [26] were selected as they represent well-established benchmarks with sufficient complexity while enabling efficient experimentation and direct comparison with prior quaternion networks. These datasets consist of 60,000 32×32 color images divided into 50,000 training and 10,000 testing samples. CIFAR-10 contains 10 classes with 6,000 images per class, providing a balanced dataset for basic object recognition. CIFAR-100 extends this to 100 classes with 600 images per class, presenting a more challenging fine-grained classification task. The relatively small image size (32×32) allows for rapid prototyping while still requiring models to develop meaningful feature representations. While larger datasets like ImageNet would provide additional validation, prior quaternion network research has primarily focused on CIFAR datasets, making them ideal for direct comparison with the state-of-the-art in quaternion networks.

2) Robotic Workspace Dataset: For oriented object detection, a custom dataset specifically designed to evaluate rotation equivariance in a realistic robotic assembly environment was created. This dataset consists of 1900 bird's-eye view images (640×640 pixels) of a UR5 robot workspace containing cardboard boxes with varying orientations, positions, and partial occlusions. Images were captured using a calibrated RGB camera mounted above the workspace, with controlled lighting conditions typical of industrial environments. Each image contains cardboard boxes of different sizes arranged in configurations representing typical assembly scenarios. Ground truth annotations include oriented bounding boxes with precise orientation angles, manually labeled and verified. The dataset was specifically designed to evaluate rotation-aware object detection in conditions relevant to robotic manipulation tasks.

B. Network Architectures

For image classification, Quaternion Wide ResNet [27] (QWideResNet) was implemented with the 16-4 configuration (16 layers, width factor 4). This architecture was selected for its efficiency and demonstrated success on CIFAR datasets. All convolutional layers were replaced with their quaternion counterparts as described in Section III. For oriented object detection, YOLOv11-nano [1] was renovated with QUAN, by replacing standard blocks with quaternion equivalents:

- C3k2 → QC3k2 (Quaternion Cross Stage Partial)
- SPPF → QSPPF (Quaternion Spatial Pyramid Pooling Fast)
- C2PSA → QC2PSA (Quaternion Partial Spatial Attention)

YOLOv11 was chosen as the base architecture due to its state-of-the-art performance in real-time object detection and its modular design, which facilitated the systematic replacement of components with quaternion equivalents. The training protocol is presented in Appendix VI-B.

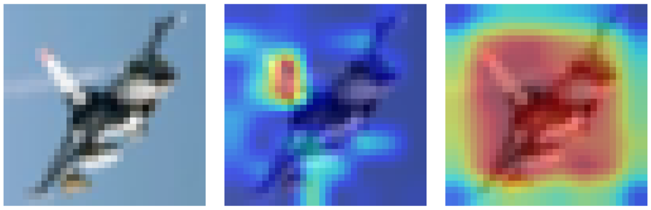

Fig. 4. (left) CIFAR-10 input Image, (middle) GradCAM visualization via real-valued, (right) visualization via quaternion-valued ResNet

C. Image Classification Performance

Table I presents the classification accuracy and parameter count for QUAN compared to both standard CNNs and existing quaternion approaches on CIFAR-10 and CIFAR-100 datasets.

TABLE I
IMAGE CLASSIFICATION ACCURACY FOR QUATERNION VALUED AND REAL-VALUED NETWORKS

Architecture	No. Params	CIFAR-10	CIFAR-100
Real-Valued CNN	3.6M	93.18%	71.93%
QUAN	**717K**	**95.12%**	**74.28%**
Deep QCNN [12]	932.8K	94.56%	73.99%
QVGG11 [25]	3.8M	85.15%	-
Shallow QCNN [14]	-	77.78%	-
QResNet34 [28]	1M	92-94%	-

The results demonstrate that QUAN achieves state-of-the-art accuracy for quaternion networks while requiring significantly fewer parameters. The QWRN-16-4 model (717K parameters) outperforms comparable quaternion networks with 23% fewer parameters than the previous best [12]. This parameter efficiency becomes even more pronounced at larger scales, where a Quaternion ResNet34 requires only 2.99M parameters compared to 21.8M for a standard ResNet34, representing an 86% reduction without sacrificing accuracy. This efficiency is particularly valuable for resource-constrained robotic systems, where model size directly impacts deployment feasibility. Despite the dramatic parameter reduction, training time increased by 6.8% on average across our experiments, confirming that QUAN's approximation approach maintains computational efficiency. Gradient-weighted Class Activation Mapping (Grad-CAM) [29] visualizations were generated to compare what parts of the image a real-valued and quaternion-valued wide ResNet look at when making predictions. In Figure 4, it is observed that the quaternion-valued model appears to generate activation maps that more precisely localize the key features of the airplane.

D. Object Detection Performance

Table II compares QUAN-YOLOv11n with standard YOLOv11n on both classical (axis-aligned) and oriented object detection tasks.

For classical object detection, QUAN-YOLOv11n outperforms standard YOLOv11n (0.9533 vs 0.915 mAP50-95) while using only 25.8% of the parameters and 36% more inference time . For oriented detection, QUAN-YOLOv11n shows a slight disadvantage in mAP50-95 (0.9102 vs. 0.9119) though with comparable mAP50. Figure 5 demonstrates QUAN's detection capabilities on the robotic workspace dataset, showing accurate oriented bounding box annotations. The slight increase in inference time (2.87ms vs. 2.098ms) is offset by the significantly smaller model size, making QUAN-YOLOv11n suitable for deployment on resource-constrained robotic systems.

TABLE II
OBJECT DETECTION PERFORMANCE

Task	Model	Params	mAP50	mAP50-95	ms
Classical	QUAN-YOLO	**0.67M**	0.9875	**0.933**	2.87
	YOLOv11n	2.61M	**0.992**	0.915	**2.098**
Oriented	QUAN-YOLOn	**0.692M**	0.9508	0.9102	4.1
	YOLOv11n	2.65M	**0.9598**	**0.9119**	**3.57**

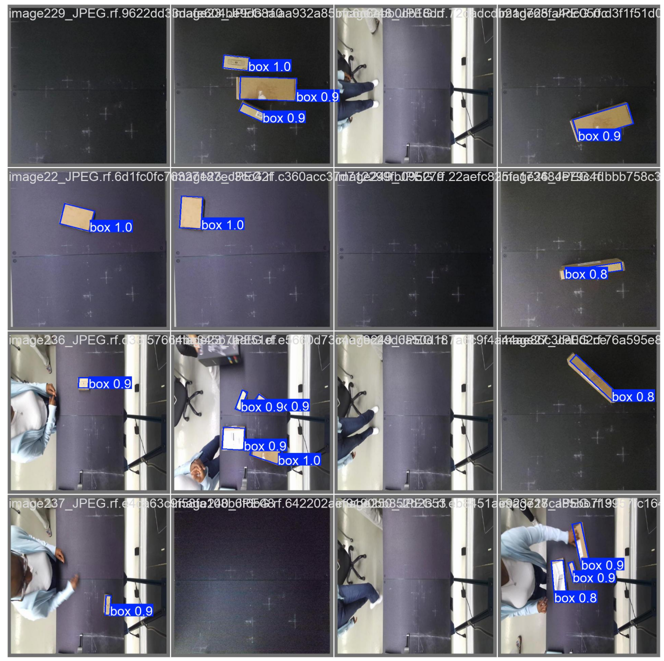

Fig. 5. Oriented Object Detection Predictions by QUAN-YOLOv11n

E. Ablation Studies

Figure 6 presents the ablation study results of quaternion activation and mapping strategies. While Poincaré mapping generally outperforms alternatives across activation functions, SiLU consistently shows superior results over ReLU and QPReLU variants. Traditional quaternion batch normalization showed on average 0.61% less accuracy than IQBN across the CIFAR-10 experiments and was not included, but may offer advantages in other contexts or with further optimization.

V. CONCLUSIONS

Quaternion Approximate Networks (QUAN) advance quaternion neural networks while maintaining practical applicability across both image classification and oriented

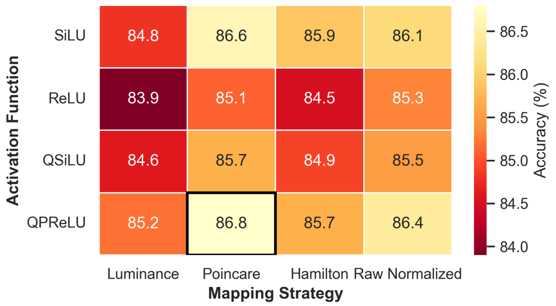

Fig. 6. Ablation studies for quaternion activation and mapping

object detection. By implementing quaternion operations through efficient real-valued approximations and independent normalization, QUAN overcomes key limitations that have hindered the widespread adoption of quaternion networks. The systematic adaptation of standard YOLO architectural blocks (C3k2, SPPF, C2PSA) to quaternion counterparts enables seamless integration of rotation equivariance into existing object detection frameworks. Notably, oriented object detection plays a crucial role in robotic assembly applications, where precise spatial understanding is essential for accurate manipulation.

Future research directions include optimizing QUAN with hardware-accelerated frameworks like JAX and scaling to larger datasets such as COCO [30], DOTA [31], and DIOR [32]. The integration of QUAN with relational keypoint approaches [33] presents a promising avenue to enhance generalization in robotic tasks through geometric constraints, potentially improving manipulation system robustness. Further exploration of quaternion-based foundation models could extend the applicability of rotation equivariant representations across various robotic perception tasks. By addressing computational challenges that have constrained quaternion networks, QUAN opens new possibilities for rotation-aware deep learning in resource-limited robotics applications, where orientation awareness is crucial for effective interaction with the physical world.

References

[1] R. Khanam and M. Hussain, "Yolov11: An overview of the key architectural enhancements," 2024. [Online]. Available: https://arxiv.org/abs/2410.17725

[2] S. Ren, K. He, R. Girshick, and J. Sun, "Faster r-cnn: Towards real-time object detection with region proposal networks," 2016. [Online]. Available: https://arxiv.org/abs/1506.01497

[3] N. Carion, F. Massa, G. Synnaeve, N. Usunier, A. Kirillov, and S. Zagoruyko, "End-to-end object detection with transformers," 2020. [Online]. Available: https://arxiv.org/abs/2005.12872

[4] C. Wang, D. Xu, Y. Zhu, R. Martín-Martín, C. Lu, L. Fei-Fei, and S. Savarese, "Densefusion: 6d object pose estimation by iterative dense fusion," 2019. [Online]. Available: https://arxiv.org/abs/1901.04780

[5] M. Caron, H. Touvron, I. Misra, H. Jégou, J. Mairal, P. Bojanowski, and A. Joulin, "Emerging properties in self-supervised vision transformers," 2021. [Online]. Available: https://arxiv.org/abs/2104.14294

[6] T. Chen, S. Kornblith, M. Norouzi, and G. Hinton, "A simple framework for contrastive learning of visual representations," 2020. [Online]. Available: https://arxiv.org/abs/2002.05709

[7] W. Fedus, B. Zoph, and N. Shazeer, "Switch transformers: Scaling to trillion parameter models with simple and efficient sparsity," 2022. [Online]. Available: https://arxiv.org/abs/2101.03961

[8] X. Yang, J. Yan, Z. Feng, and T. He, "R3det: Refined single-stage detector with feature refinement for rotating object," 2020. [Online]. Available: https://arxiv.org/abs/1908.05612

[9] Y. Xu, M. Fu, Q. Wang, Y. Wang, K. Chen, G.-S. Xia, and X. Bai, "Gliding vertex on the horizontal bounding box for multi-oriented object detection," *IEEE Transactions on Pattern Analysis and Machine Intelligence*, vol. 43, no. 4, p. 1452–1459, Apr. 2021. [Online]. Available: http://dx.doi.org/10.1109/TPAMI.2020.2974745

[10] W. R. Hamilton, *Elements of quaternions*. BoD–Books on Demand, 2022.

[11] J. B. Kuipers, "Quaternion algebra," in *Quaternions and Rotation Sequences: A Primer With Applications to Orbits, Aerospace and Virtual Reality*. Princeton Univ. Press, 1999, p. 125.

[12] C. Gaudet and A. Maida, "Deep quaternion networks," 2018. [Online]. Available: https://arxiv.org/abs/1712.04604

[13] T. Parcollet, M. Ravanelli, M. Morchid, G. Linarès, C. Trabelsi, R. D. Mori, and Y. Bengio, "Quaternion recurrent neural networks," 2019. [Online]. Available: https://arxiv.org/abs/1806.04418

[14] X. Zhu, Y. Xu, H. Xu, and C. Chen, "Quaternion convolutional neural networks," 2019. [Online]. Available: https://arxiv.org/abs/1903.00658

[15] G. Altamirano-Gomez and C. Gershenson, "Quaternion convolutional neural networks: Current advances and future directions," 2023. [Online]. Available: https://arxiv.org/abs/2307.08663

[16] T. Hodan, F. Michel, E. Brachmann, W. Kehl, A. G. Buch, D. Kraft, B. Drost, J. Vidal, S. Ihrke, X. Zabulis, C. Sahin, F. Manhardt, F. Tombari, T.-K. Kim, J. Matas, and C. Rother, "Bop: Benchmark for 6d object pose estimation," 2018. [Online]. Available: https://arxiv.org/abs/1808.08319

[17] K. Kleeberger, C. Landgraf, and M. F. Huber, "Large-scale 6d object pose estimation dataset for industrial bin-picking," 2019. [Online]. Available: https://arxiv.org/abs/1912.12125

[18] D. Comminiello, M. Lella, S. Scardapane, and A. Uncini, "Quaternion convolutional neural networks for detection and localization of 3d sound events," in *ICASSP 2019 - 2019 IEEE International Conference on Acoustics, Speech and Signal Processing (ICASSP)*, 2019, pp. 8533–8537.

[19] E. Grassucci, D. Comminiello, and A. Uncini, "A quaternion-valued variational autoencoder," in *ICASSP 2021 - 2021 IEEE International Conference on Acoustics, Speech and Signal Processing (ICASSP)*. IEEE, June 2021. [Online]. Available: http://dx.doi.org/10.1109/ICASSP39728.2021.9413859

[20] H.-W. Hsu, T.-Y. Wu, S. Wan, W. H. Wong, and C.-Y. Lee, "Quatnet: Quaternion-based head pose estimation with multiregression loss," *IEEE Transactions on Multimedia*, vol. 21, no. 4, pp. 1035–1046, 2019.

[21] Y. He, W. Sun, H. Huang, J. Liu, H. Fan, and J. Sun, "PVN3D: A deep point-wise 3d keypoints voting network for 6dof pose estimation," *CoRR*, vol. abs/1911.04231, 2019. [Online]. Available: http://arxiv.org/abs/1911.04231

[22] W. Li, Z. Yang, W. Han, H. Man, X. Wang, and X. Fan, "Hyperbolic-constraint point cloud reconstruction from single rgb-d images," 2024. [Online]. Available: https://arxiv.org/abs/2412.09055

[23] M. Nickel and D. Kiela, "Poincaré embeddings for learning hierarchical representations," 2017. [Online]. Available: https://arxiv.org/abs/1705.08039

[24] H. Zhou, X. Zhang, C. Zhang, and Q. Ma, "Quaternion convolutional neural networks for hyperspectral image classification," *Engineering Applications of Artificial Intelligence*, vol. 123, p. 106234, 2023. [Online]. Available: https://www.sciencedirect.com/science/article/pii/S0952197623004189

[25] J. Pöppelbaum and A. Schwung, "Improving quaternion neural networks with quaternionic activation functions," 2024. [Online]. Available: https://arxiv.org/abs/2406.16481

[26] A. Krizhevsky and G. Hinton, "Learning multiple layers of features from tiny images," University of Toronto, Toronto, Ontario, Tech. Rep. 0, 2009. [Online]. Available: https://www.cs.toronto.edu/~kriz/learning-features-2009-TR.pdf

[27] S. Zagoruyko and N. Komodakis, "Wide residual networks," 2017. [Online]. Available: https://arxiv.org/abs/1605.07146

[28] S. Hongo, T. Isokawa, N. Matsui, H. Nishimura, and N. Kamiura, "Constructing convolutional neural networks based on quaternion," in *2020 International Joint Conference on Neural Networks (IJCNN)*, 2020, pp. 1–6.

TABLE III

QUAN TRAINING PROTOCOL

Task	Parameter	Value
Classification	Batch size	256
	Initial LR	0.1
	Momentum	0.9
	Weight decay	1e-4
	LR schedule	10× drop at 75, 150
	Epochs	300
	Augmentation	Random crops, h-flips, AutoAug
Object Detection	Batch size	32
	Optimizer	AdamW w/ auto param tuning
	Image size	640×640
	Data workers	12
	Augmentation	Mosaic, affine transforms, h-flips
	Epochs	100

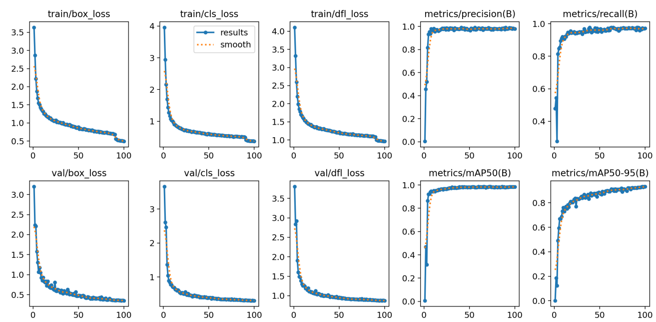

Fig. 7. Classical Object Detection Results by QUAN-YOLOv11n

[29] R. R. Selvaraju, M. Cogswell, A. Das, R. Vedantam, D. Parikh, and D. Batra, "Grad-cam: Visual explanations from deep networks via gradient-based localization," *International Journal of Computer Vision*, vol. 128, no. 2, p. 336–359, Oct. 2019. [Online]. Available: http://dx.doi.org/10.1007/s11263-019-01228-7

[30] T.-Y. Lin, M. Maire, S. Belongie, L. Bourdev, R. Girshick, J. Hays, P. Perona, D. Ramanan, C. L. Zitnick, and P. Dollár, "Microsoft coco: Common objects in context," 2015. [Online]. Available: https://arxiv.org/abs/1405.0312

[31] G.-S. Xia, X. Bai, J. Ding, Z. Zhu, S. Belongie, J. Luo, M. Datcu, M. Pelillo, and L. Zhang, "Dota: A large-scale dataset for object detection in aerial images," 2019. [Online]. Available: https://arxiv.org/abs/1711.10398

[32] K. Li, G. Wan, G. Cheng, L. Meng, and J. Han, "Object detection in optical remote sensing images: A survey and a new benchmark," *ISPRS Journal of Photogrammetry and Remote Sensing*, vol. 159, p. 296–307, Jan. 2020. [Online]. Available: http://dx.doi.org/10.1016/j.isprsjprs.2019.11.023

[33] W. Huang, C. Wang, Y. Li, R. Zhang, and L. Fei-Fei, "Rekep: Spatio-temporal reasoning of relational keypoint constraints for robotic manipulation," 2024. [Online]. Available: https://arxiv.org/abs/2409.01652

VI. APPENDIX

A. Mapping Strategies

Multiple RGB-to-quaternion mapping strategies have been investigated to determine the optimal approach for preserving both color relationships and geometric properties. The evaluated mappings included:

- **Luminance mapping**: Uses perceptual weights (0.299R + 0.587G + 0.114B) for the real component while preserving normalized RGB values in the imaginary components, providing human perception-aligned representations.
- **Mean brightness mapping**: Utilizes the arithmetic mean of RGB channels as the real component, offering computational simplicity.
- **Raw normalized mapping**: Directly normalizes RGB values with equal weighting across channels.
- **Hamilton mapping**: Embeds RGB values as pure imaginary components (0, R, G, B), preserving raw color information.
- **Poincaré mapping**: Projects RGB values onto the unit quaternion sphere, ensuring numerical stability during training while preserving relative color relationships.

B. Training Protocol

The training protocol for the experiments can be found in Table III.

C. Object Detection Graphs

The QUAN-YOLOv11n model's classical detection performance on the robotic workspace dataset is summarized in Figure 7 while the oriented detection performance is summarized in Figure 8. The quaternion-based approach achieves comparable mAP to standard YOLOv11n while using only 26% percent of the original model parameters.

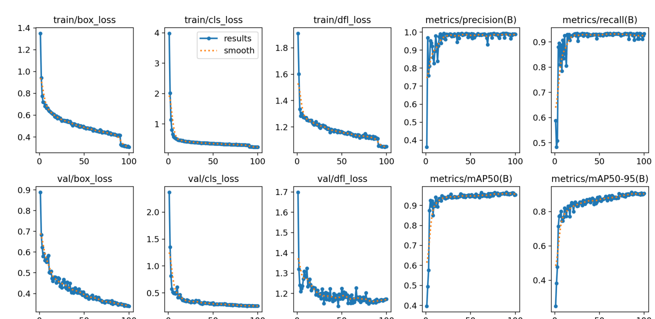

Fig. 8. Oriented Object Detection by QUAN-YOLOv11n